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MIL-M-28787/524
26 November 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

NATO-STD-4156 (MODIFIED) CHANNEL ADAPTER MODULE, KEY CODE ATL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NATO-STD-4156 (modified) channel adapter module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.638 failures per million operating hours for class I or 3.699 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be ATL.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A45	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Power Supply Stress	IV	Q	----------Module shall pass all remaining tests----------						
D1	Ibd	V	S	-0.02	2.42	-0.02	2.42	-0.02	2.42	
D2,D3	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	
D4	Ibd	V	S	0.0	2.1	0.0	2.1	0.0	2.1	
D5	Ibd	V	S	0.0	3.2	0.0	3.2	0.0	3.2	
D6	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	
D7	Ibd	V	S	0.0	3.3	0.0	3.3	0.0	3.3	
D8	Ibd	V	S	-0.02	1.12	-0.02	1.12	-0.02	1.12	
D9	Ibd	V	S	-0.02	2.12	-0.02	2.12	-0.02	2.12	
D10	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	
D11	Ibd	V	S	-0.02	1.02	-0.02	1.02	-0.02	1.02	
D12,D13	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	
D14	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	
D15,D16	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	
E1-E3	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	
E4-E9	Ios	VI	S	-150.0	-30.0	-150.0	-30.0	-150.0	-30.0	
E10,E11	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	
E12	Ioe	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	
E13-E16	Ioe	VI	S	-150.0	-30.0	-150.0	-30.0	-150.0	-30.0	

- See footnotes at end of table.

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TABLE 1. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class 1 0 ± 5°C 55 ± 5°C 4/		Class 1A -55 ± 5°C 85 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
F1	11h	VII	x	-7.0	-0.46	-7.0	-0.46	-7.0	-0.46	µA
F2,F3	11h	VII	x	-0.05	0.05	-0.05	0.05	-0.05	0.05	µA
F4	11h	VII	x	-2.05	0.1	-2.05	0.1	-2.05	0.1	µA
F5	11h	VII	x	-4.9	0.14	-4.9	0.14	-4.9	0.14	µA
F8	11h	VII	x	-2.25	0.08	-2.25	0.08	-2.25	0.08	µA
F7-F10	11h	VII	x	-2.0	2.3	-2.0	2.3	-2.0	2.3	µA
F11	11h	VII	x	-5.5	0.16	-5.5	0.16	-5.5	0.16	µA
F12	11h	VII	x	-2.00	-0.57	-2.00	-0.57	-2.00	-0.57	µA
F13	11h	VII	x	-4.00	-0.52	-4.00	-0.52	-4.00	-0.52	µA
F14	11h	VII	x	-2.04	-0.53	-2.04	-0.53	-2.04	-0.53	µA
F15	11h	VII	x	-0.25	0.03	-0.25	0.03	-0.25	0.03	µA
F16	11h	VII	x	-3.15	-0.38	-3.15	-0.38	-3.15	-0.38	µA
F17,F18	11h	VII	x	-0.5	0.02	-0.5	0.02	-0.5	0.02	µA
F19	11h	VII	x	-1.2	0.04	-1.2	0.04	-1.2	0.04	µA
F20	11h	VII	x	-0.25	0.03	-0.25	0.03	-0.25	0.03	µA
F21,F22	11h	VII	x	-2.0	2.3	-2.0	2.3	-2.0	2.3	µA
F23	11h	VII	x	-2.00	0.05	-2.00	0.05	-2.00	0.05	µA
F24-F27	11h	VII	x	-2.0	2.3	-2.0	2.3	-2.0	2.3	µA
G1	11i	VIII	x	-7.0	-0.46	-7.0	-0.46	-7.0	-0.46	µA
G2,G3	11i	VIII	x	-0.05	0.05	-0.05	0.05	-0.05	0.05	µA
G4	11i	VIII	x	-2.05	0.1	-2.05	0.1	-2.05	0.1	µA
G5	11i	VIII	x	-4.90	0.14	-4.90	0.14	-4.90	0.14	µA
G6	11i	VIII	x	-2.25	0.08	-2.25	0.08	-2.25	0.08	µA
G7-G10	11i	VIII	x	-2.0	2.3	-2.0	2.3	-2.0	2.3	µA
G11	11i	VIII	x	-5.5	0.16	-5.5	0.16	-5.5	0.16	µA
G12	11i	VIII	x	-2.00	-0.57	-2.00	-0.57	-2.00	-0.57	µA
G13	11i	VIII	x	-4.0	-0.52	-4.0	-0.52	-4.0	-0.52	µA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 80 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G14	Iil	VIII	X	-2.84	-0.53	-2.84	-0.53	-2.84	-0.53	μA
G15	Iil	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	μA
G16	Iil	VIII	X	-3.15	-0.56	-3.15	-0.56	-3.15	-0.56	μA
G17,G18	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	μA
G19	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	μA
G20	Iil	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	μA
G21,G22	Iil	VIII	X	-2.8	2.3	-2.8	2.3	-2.8	2.3	μA
G23	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	μA
G24-G27	Iil	VIII	X	-2.8	2.3	-2.8	2.3	-2.8	2.3	μA
H1-H15	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J18	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	μA
K2	Iozh	XI	X	-0.65	0.30	-0.65	0.30	-0.65	0.30	μA
K3-K5	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	μA
K6	Iozh	XI	X	-0.65	0.30	-0.65	0.30	-0.65	0.30	μA
K7-K10	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	μA
L1,L2	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L3-L6	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L7	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L8-L10	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L11	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L12	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
M1	PSFC	XIII	S	8.4	15.6	8.4	15.6	8.4	15.6	uF
M2	PSFC	XIII	S	5.6	10.4	5.6	10.4	5.6	10.4	uF
N	Icc	XIV	X	725	1740	725	1740	725	1740	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	Cin	XVIII	Q	14	200	14	200	14	200	pF
R2	Cin	XVIII	Q	4	175	4	175	4	175	pF
R3	Cin	XVIII	Q	4	150	4	150	4	150	pF
R4	Cin	XVIII	Q	6	125	6	125	6	125	pF
R5	Cin	XVIII	Q	10	200	10	200	10	200	pF
R6	Cin	XVIII	Q	4	100	4	100	4	100	pF
R7-R9	Cin	XVIII	Q	2	100	2	100	2	100	pF
R10	Cin	XVIII	Q	2	75	2	75	2	75	pF
R11	Cin	XVIII	Q	12	175	12	175	12	175	pF
R12	Cin	XVIII	Q	2	125	2	125	2	125	pF
R13	Cin	XVIII	Q	6	125	6	125	6	125	pF
R14	Cin	XVIII	Q	9	200	9	200	9	200	pF
R15	Cin	XVIII	Q	2	125	2	125	2	125	pF
R16	Cin	XVIII	Q	6	100	6	100	6	100	pF
R17,R18	Cin	XVIII	Q	2	75	2	75	2	75	pF
R19,R20	Cin	XVIII	Q	2	100	2	100	2	100	pF
R21-R27	Cin	XVIII	Q	2	75	2	75	2	75	pF
S	Vic Latchup	XIX	Q	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1,T2	PSTR	XX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/524-1 for class I and M28787/524-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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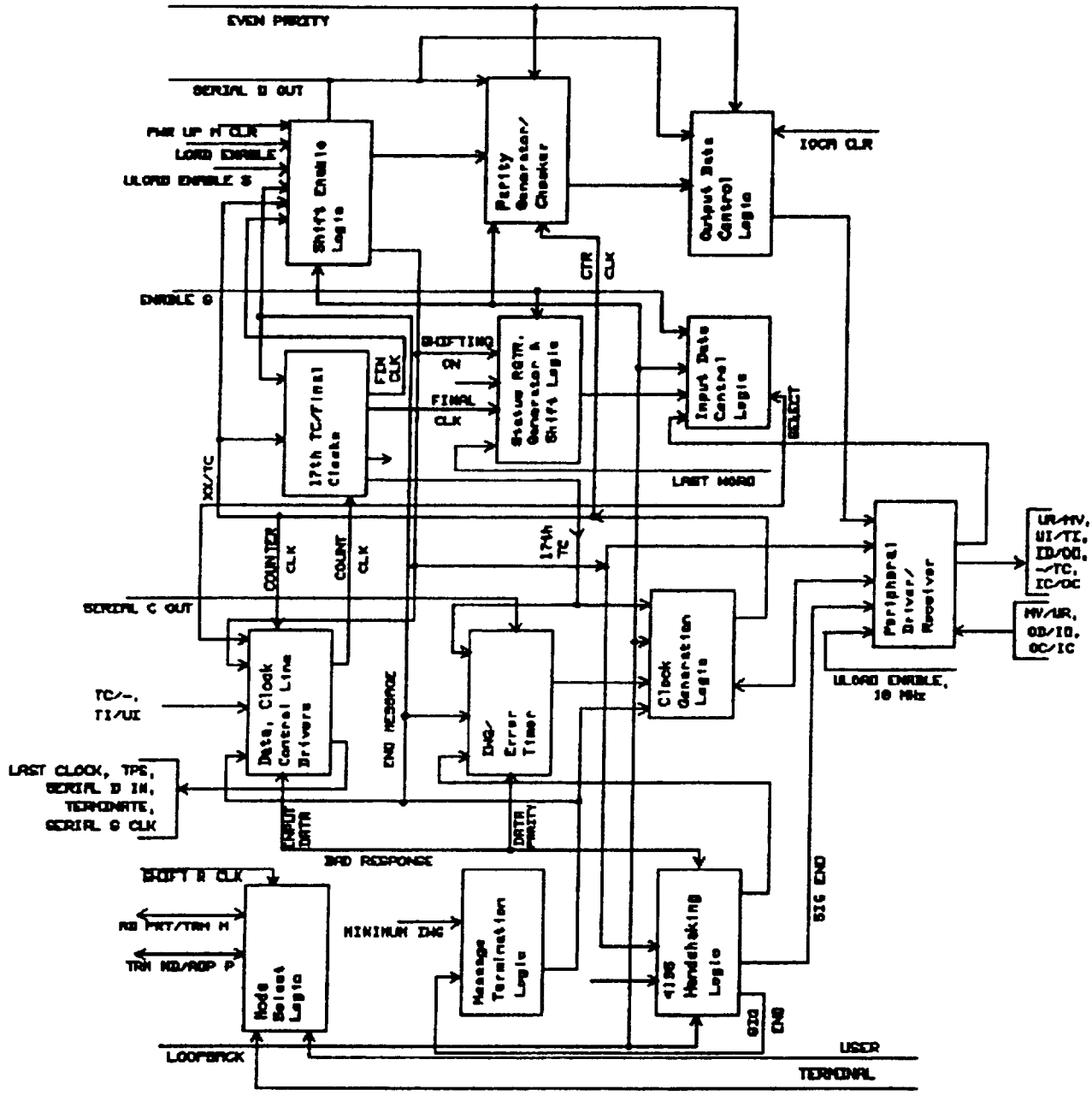


FIGURE 1. ATL functional block diagram.

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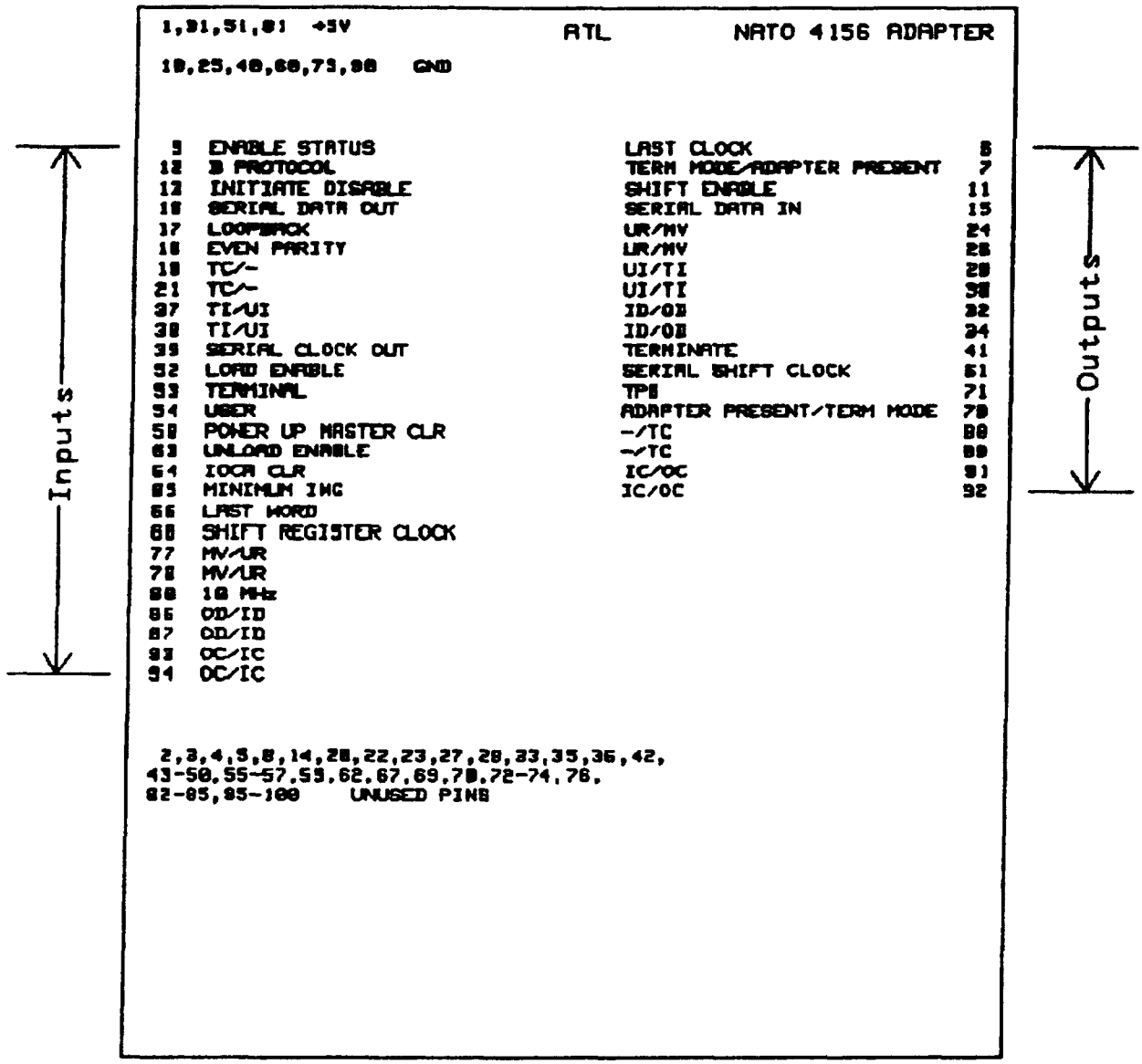


FIGURE 2. ATL pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L and S are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, L or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L and S where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

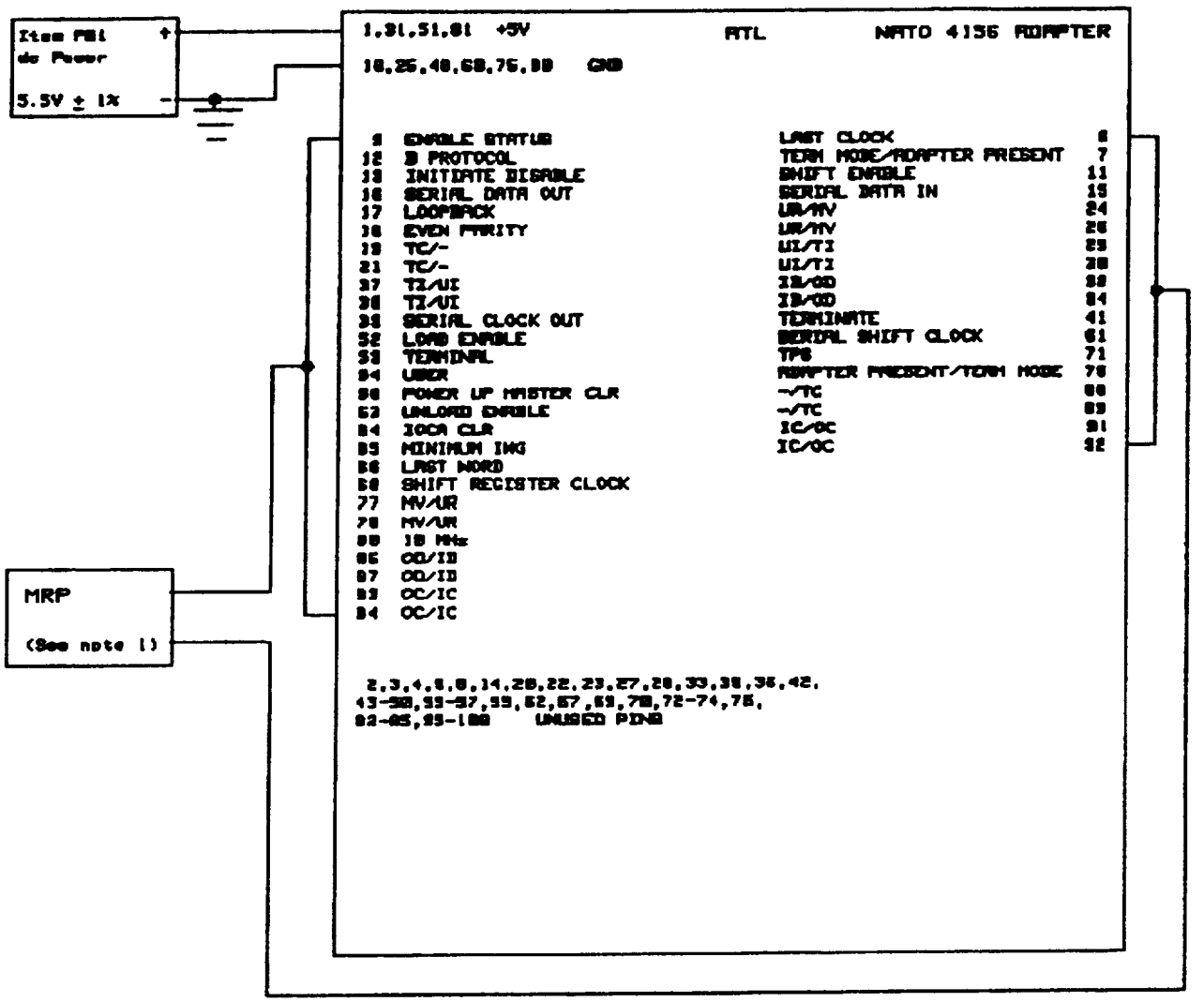
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. ATL operating temperature and component temperature test circuit.

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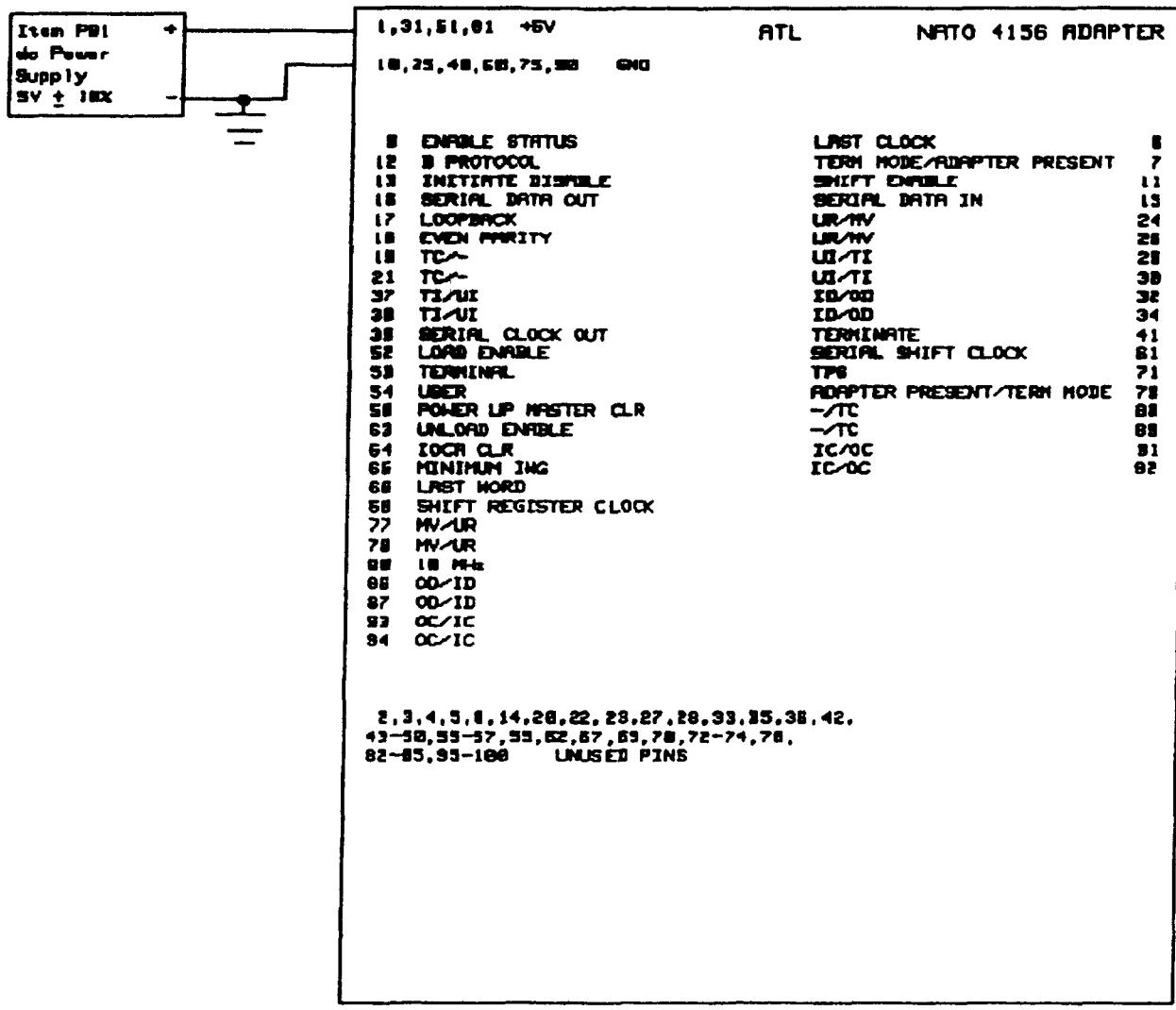


FIGURE 4. ATL vibration test circuit.

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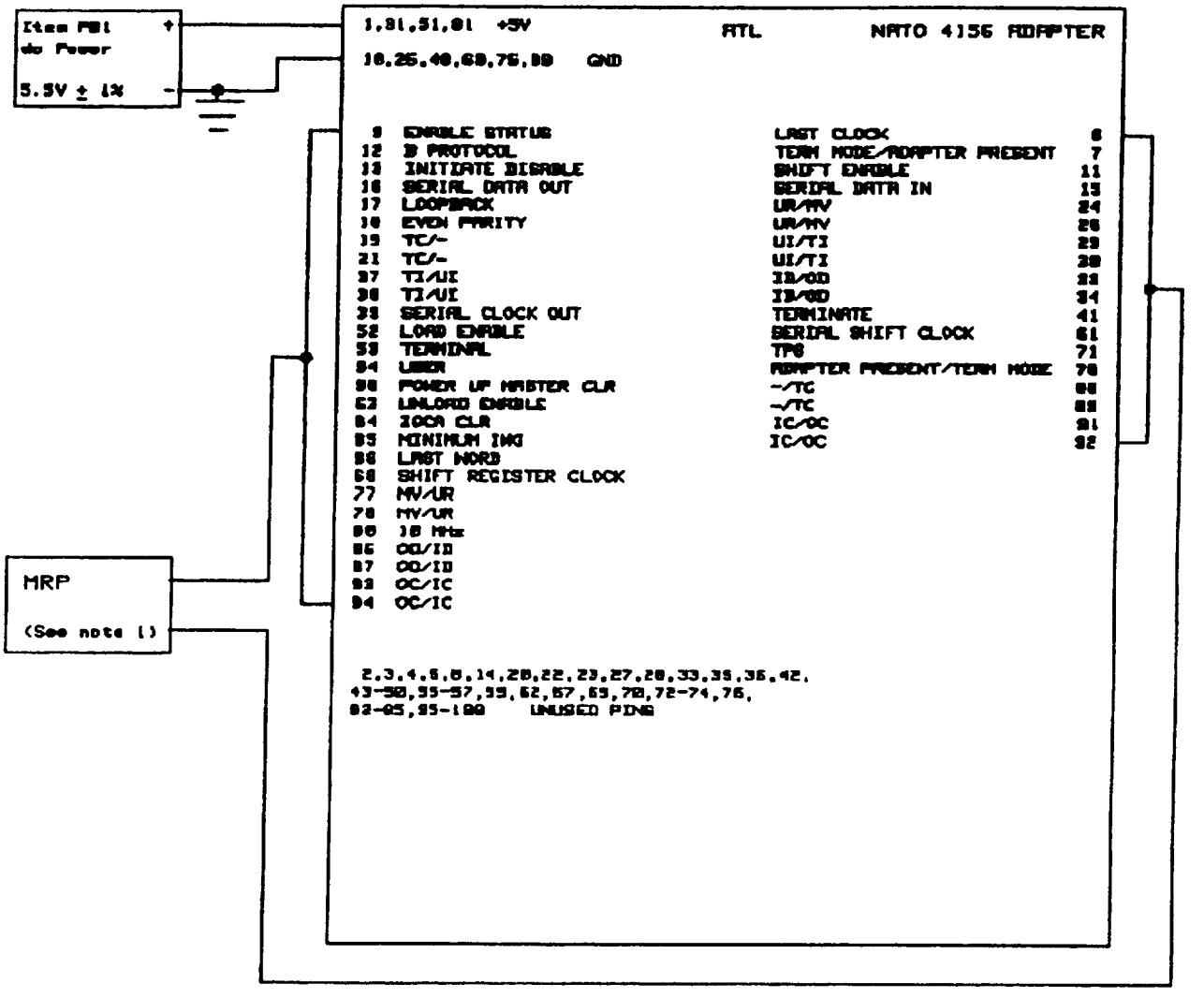
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. ATL life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ibd = input breakdown current.
(c) Icc = power supply current.
(d) Iih = input high level current.
(e) Iil = input low level current.
(f) Iin = input forcing current.
(g) Ioh = output high level current.
(h) Iol = output low level current.
(i) Ios = output short circuit current.
(j) Iozh = output high level leakage current.
(k) Iozl = output low level leakage current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Vin = input forcing voltage.
(s) Voh = output high level voltage.
(t) Vol = output low level voltage.
(u) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 9.57 W (max); 6.41 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ENABLE S	NA	0.4	2.4	-7.0	-0.46	2.42	200
B PROTOCOL	-1.5	0.4	2.4	-0.85	0.05	1.1	175
INIT DISABLE	-1.5	0.4	2.4	-0.85	0.05	1.1	150
SERIAL D OUT	-1.2	0.4	2.4	-2.85	0.1	2.1	125
LOOPBACK	-1.2	0.4	2.4	-4.9	0.14	3.2	200
EVEN PARITY	-1.2	0.4	2.4	-2.25	0.08	2.0	100
TC/- (+)	NA	0.4	2.4	-2.8	2.3	NA	100
TC/- (-)	NA	0.4	2.4	-2.8	2.3	NA	100
TI/UI (-)	NA	0.4	2.4	-2.8	2.3	NA	100
TI/UI (+)	NA	0.4	2.4	-2.8	2.3	NA	75
SERIAL C OUT	-1.2	0.4	2.4	-5.5	0.16	3.3	175
LOAD ENABLE	NA	0.4	2.4	-2.0	-0.57	1.12	125
TERMINAL	NA	0.4	2.4	-4.0	-0.52	2.12	125
USER	NA	0.4	2.4	-2.04	-0.53	NA	200

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PWR UP M CLR	-1.5	0.4	2.4	-0.25	0.03	1.0	125
ULOAD ENABLE	NA	0.4	2.4	-3.15	-0.56	1.02	100
IOCA CLR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MINIMUM IWG	-1.2	0.4	2.4	-0.6	0.02	0.1	75
LAST WORD	-1.2	0.4	2.4	-1.2	0.04	0.2	100
SHIFT R CLK	-1.5	0.4	2.4	-0.25	0.03	1.0	100
MV/UR (-)	NA	0.4	2.4	-2.8	2.3	NA	75
MV/UR (+)	NA	0.4	2.4	-2.8	2.3	NA	75
10 MHz	-1.2	0.4	2.4	-2.0	0.05	1.0	75
OD/ID (+)	NA	0.4	2.4	-2.8	2.3	NA	75
OD/ID (-)	NA	0.4	2.4	-2.8	2.3	NA	75
OC/IC (+)	NA	0.4	2.4	-2.8	2.3	NA	75
OC/IC (-)	NA	0.4	2.4	-2.8	2.3	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
LAST CLOCK	2.4	0.5	-0.4	6.0	0.05	0.05	-50.0	NA
TRM MD/ADP P	NA	0.5	NA	1.6	0.30	NA	NA	NA
SHIFT ENABLE	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
SERIAL D IN	2.4	0.5	-0.4	6.0	0.05	0.05	-50.0	NA
UR/MV (L)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
UR/MV (H)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
UI/TI (L)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
UI/TI (H)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
ID/OD (L)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
ID/OD (H)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
TERMINATE	2.4	0.5	-0.4	6.0	0.05	0.05	-50.0	NA
SERIAL S CLK	2.4	0.5	-0.4	6.0	0.05	0.05	-50.0	NA
TP6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
AD PRT/TRM M	NA	0.5	NA	1.6	0.30	NA	NA	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
-/TC (L)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
-/TC (H)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
IC/OC (L)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
IC/OC (H)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	OC/IC (+) to SHIFT ENABLE (low level)	-	-	-	212	212	212	nS	11
1	tpd	OC/IC (-) to SHIFT ENABLE (low level)	-	-	-	212	212	212	nS	11
2	tpd	TI/UI (-) to SHIFT ENABLE (high level)	-	-	-	327	327	327	nS	11
2	tpd	TI/UI (+) to SHIFT ENABLE (high level)	-	-	-	327	327	327	nS	11
3	tpd	MV/UR (-) to TERMINATE	-	-	-	256	256	256	nS	11
3	tpd	MV/UR (+) to TERMINATE	-	-	-	256	256	256	nS	11
4	tpd	OC/IC (+) to SERIAL S CLK	-	-	-	109	109	109	nS	11
4	tpd	OC/IC (-)to SERIAL S CLK	-	-	-	109	109	109	nS	11
5	tpd	OD/ID (+) to SERIAL D IN	-	-	-	127	127	127	nS	11
5	tpd	OD/ID (-) to SERIAL D IN	-	-	-	127	127	127	nS	11
6	tpd	TC/- (+) to LAST CLOCK	-	-	-	127	127	127	nS	11
6	tpd	TC/- (-) to LAST CLOCK	-	-	-	127	127	127	nS	11
7	tpd	MV/UR (-) to UI/TI (L)	-	-	-	211	211	211	nS	11
7	tpd	MV/UR (-) to UI/TI (H)	-	-	-	211	211	211	nS	11
7	tpd	MV/UR (+) to UI/TI (L)	-	-	-	211	211	211	nS	11
7	tpd	MV/UR (+) to UI/TI (H)	-	-	-	211	211	211	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	ULOAD ENABLE to UR/MV (L)	-	-	-	131	131	131	nS	11
8	tpd	ULOAD ENABLE to UR/MV (H)	-	-	-	131	131	131	nS	11
9	tpd	SHIFT R CLK to IC/OC (L)	-	-	-	94	94	94	nS	11
9	tpd	SHIFT R CLK to IC/OC (H)	-	-	-	94	94	94	nS	11
10	tpd	SERIAL C OUT to -/TC (L)	-	-	-	82	82	82	nS	11
10	tpd	SERIAL C OUT to -/TC (H)	-	-	-	82	82	82	nS	11

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6.4 Module function. The NATO-STD-4156 (modified) serial interface module contains circuits to communicate in user and terminal modes. The module contains mode select logic, data, clock and control line drivers, 17th Terminal Clock (TC)/final register, status generator and status shift clock, a parity generator/checker, 4156 handshaking logic, clock generation logic, input/output data control logic and peripheral drivers and receivers.

6.4.1 Mode select logic. The NATO-STD-4156 (modified) channel control module (ATK) selects the proper serial interface module for the communications mode required. A serial interface module is hardwire strapped to respond as a user or terminal adapter. The mode select logic is composed of four gates and part of an AND gate array. The output of the AND gate array generates the active low select signal to select the internal clock rate and determine when serial interface module usage is valid. When the mode selected does not agree with the appropriate module strapping, a class III, priority 2 interrupt is generated.

6.4.2 Data, clock and control line drivers. These drivers are used to transfer data, clock and control information to the serial channel control module. The drivers are enabled by the select signal from the mode select logic.

6.4.3 17th TC/final clocks. The 17th TC and final clocks are composed of a series of counters which count the number of clock pulses used per word, in a normal message and status transfer. The counters are cleared at the end of the status or normal message or when cleared by a power-up or Input/Output Channel Adapter (IOCA) master clear.

6.4.4 Shift enable logic. The shift enable logic is composed of a group of flip-flops and part of an AND gate array. Based on the count of the active low final clock signal, this logic puts the shift registers on this module and on this serial channel control module (ATK) into the shift mode.

6.4.5 Handshaking logic. The handshaking logic ensures the correct handshake line is set at the proper time.

6.4.6 Message termination logic. Message termination logic is used to set active low terminate signal to the serial channel control module active after the end of a message. This occurs after all handshake lines have been dropped to their inactive state. During a status type of transfer, terminate is held active during the entire transfer.

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6.4.7 IWG/error timer. In the terminal mode, the InterWord Gap (IWG)/error timer sets the interword gap to 10 clock pulses. It also stops the Militarized Reconfigurable Processor (MRP) from trying to initiate a transfer after 10 clock pulses. In the user mode, the timer prevents the MRP from responding to a terminal device if it has not done so in 3.3 microseconds. If any of these events occur, the appropriate bit is set in the serial status register.

6.4.8 Parity generator/checker. The parity generator/checker is always enabled to generate/check odd parity. An even parity condition is simulated during BIT testing in the loopback mode to test the logic. The parity generator/checker is clocked by the active high counter clock signal from the clock generation logic. The parity checker output is sent to the status register and to an AND gate array to enable the counter clock (part of the clock generation logic).

6.4.9 Status register, generator and shift logic. This group of logic contains the status shift register, the reset status logic, the logic for putting the shift register into the shift mode and logic that sets some of the status bits.

6.4.10 Output data control logic. The output data control logic selects even or odd parity (even parity is used during BIT testing in the loopback mode) and enables the parity bit to the peripheral driver on the 17th clock pulse.

6.4.11 Input data control logic. The input data control logic consists of two AND gate array multiplexers, one of which selects peripheral data or loopback data (during Built-In-Test (BIT) testing only) and the other selects parity data or status information. Outputs from the input data control logic are sent to the data clock and control line drivers.

6.4.12 Peripheral driver/receivers. The peripheral driver circuits are differential line driver circuits with complementary outputs which are used to provide unipolar drive to twisted-pair or parallel wire transmission lines. The peripheral receivers are differential line receivers which use fail-safe input/output protection to ensure the outputs are kept high when the inputs are open (i.e., no input data).

6.4.13 Clock generation logic. The clock generation logic is comprised of a status clock, a counter clock and logic to start the terminal clocks. It also has circuits to steer the serial clock out signal, the shift register clock signal, the TC/XX signal and OC/IC to the appropriate logic at the proper time.

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6.4.14 Description of I/O pins.

Signal name	Pin number	I/O	Description
ENABLE S	9	I	ENABLE STATUS
B PROTOCOL	12	I	B PROTOCOL
INIT DISABLE	13	I	INITIATE DISABLE
SERIAL D OUT	16	I	SERIAL DATA OUT
LOOPBACK	17	I	LOOPBACK
EVEN PARITY	18	I	EVEN PARITY
TC/- (+)	19	I	TC/-
TC/- (-)	21	I	TC/-
TI/UI (-)	37	I	TI/UI
TI/UI (+)	38	I	TI/UI
SERIAL C OUT	39	I	SERIAL CLOCK OUT
LOAD ENABLE	52	I	LOAD ENABLE
TERMINAL	53	I	TERMINAL
USER	54	I	USER
PWR UP M CLR	58	I	POWER UP MASTER CLR
ULOAD ENABLE	63	I	UNLOAD ENABLE
IOCA CLR	64	I	IOCA CLR
MINIMUM IWG	65	I	MINIMUM IWG
LAST WORD	66	I	LAST WORD
SHIFT R CLK	68	I	SHIFT REGISTER CLOCK

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6.4.14 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MV/UR (-)	77	I	MV/UR
MV/UR (+)	78	I	MV/UR
10 MHz	80	I	10 MHz
OD/ID (+)	86	I	OD/ID
OD/ID (-)	87	I	OD/ID
OC/IC (+)	93	I	OC/IC
OC/IC (-)	94	I	OC/IC
LAST CLOCK	6	O	LAST CLOCK
TRM MD/ADP P	7	O	TERM MODE/ADAPTER PRESENT
SHIFT ENABLE	11	O	SHIFT ENABLE
SERIAL D IN	15	O	SERIAL DATA IN
UR/MV (L)	24	O	UR/MV
UR/MV (H)	26	O	UR/MV
UI/TI (L)	29	O	UI/TI
UI/TI (H)	30	O	UI/TI
ID/OD (L)	32	O	ID/OD
ID/OD (H)	34	O	ID/OD
TERMINATE	41	O	TERMINATE
SERIAL S CLK	61	O	SERIAL SHIFT CLOCK
TP6	71	O	TP6

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6.4.14 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
AD PRT/TRM M	79	O	ADAPTER PRESENT/TERM MODE
-/TC (L)	88	O	-/TC
-/TC (H)	89	O	-/TC
IC/OC (L)	91	O	IC/OC
IC/OC (H)	92	O	IC/OC

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0071)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C, all unused pins	Frame	10	INF	MOhm	1-9
A2	Isolation, used to unused pins	C	2,3,4,5, 8,14,20, 22,23,27, 28,33,35, 36,42,43, 44,45,46, 47,48,49, 50,55,56, 57,59,62, 67,69,70, 72,73,74, 76,82,83, 84,85,95, 96,97,98, 99,100				
A3-A46	Isolation, unused to unused pins	See note 4		10	INF	MOhm	1-9

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A46
Isolation measurement	NA	± 500 KOhm	A1-A46

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. The unused pins on this module are: 2,3,4,5,8,14,20,22,23, 27,28,33,35,36,42,43,44,45,46,47,48,49,50,55,56,57,59,62,67,69, 70,72,73,74,76,82,83,84,85,95,96,97,98,99,100. When testing for unused to unused pin isolation, connect one unused pin to A and the remaining unused pins to B.

5. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

6. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

7. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

8. The test circuit shall be as indicated on figure 6.

9. Abbreviations used are defined in 6.1.

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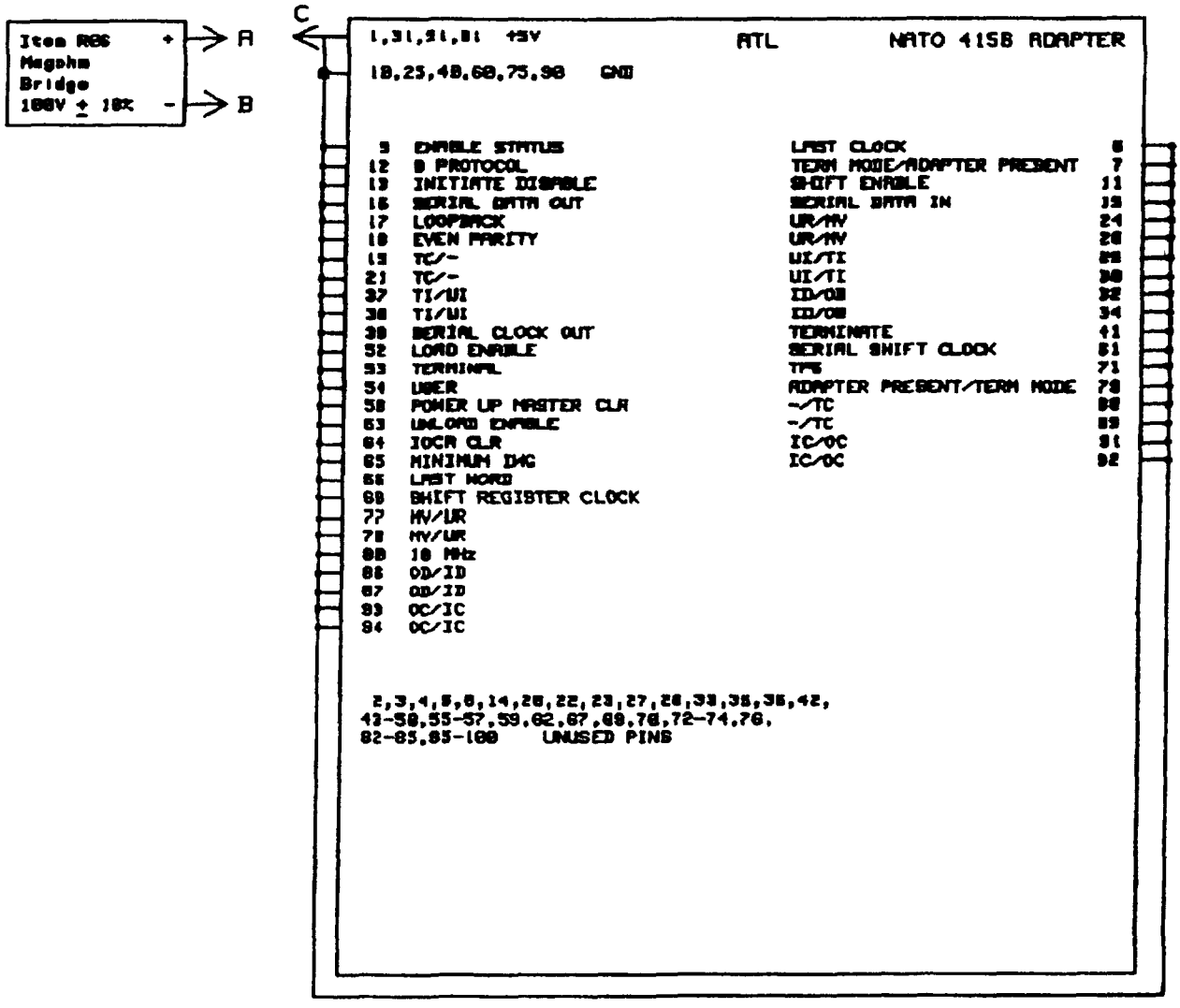


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,61,91 +8V ATL NATO 4158 ADAPTER

10,25,40,50,75,90 GND

9	ENABLE STATUS	LAST CLOCK	8
12	B PROTOCOL	TERM MODE/ADAPTER PRESENT	7
13	INITIATE DISABLE	SHIFT ENABLE	11
16	SERIAL DATA OUT	SERIAL DATA IN	15
17	LOOPBACK	UR/MV	24
18	EVEN PARITY	UR/MV	28
19	TC/-	UI/TI	29
21	TC/-	UI/TI	30
37	TI/UI	ID/OD	32
38	TI/UI	ID/OD	34
39	SERIAL CLOCK OUT	TERMINATE	41
52	LOAD ENABLE	SERIAL SHIFT CLOCK	61
53	TERMINAL	TP6	71
54	USER	ADAPTER PRESENT/TERM MODE	78
58	POWER UP MASTER CLR	-/TC	80
63	UNLOAD ENABLE	-/TC	80
64	IOCR CLR	IC/OC	81
65	MINIMUM IMG	IC/OC	82
68	LAST WORD		
69	SHIFT REGISTER CLOCK		
77	MV/UR		
78	MV/UR		
80	10 MHz		
86	OD/ID		
87	OD/ID		
93	OC/IC		
94	OC/IC		

1,3,4,5,6,14,20,22,23,27,28,33,35,36,42,
43-50,55-57,59,62,67,69,70,72-74,76,
82-85,95-100 UNUSED PINS

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	3.13 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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Check the source to verify that this is the current version before use.

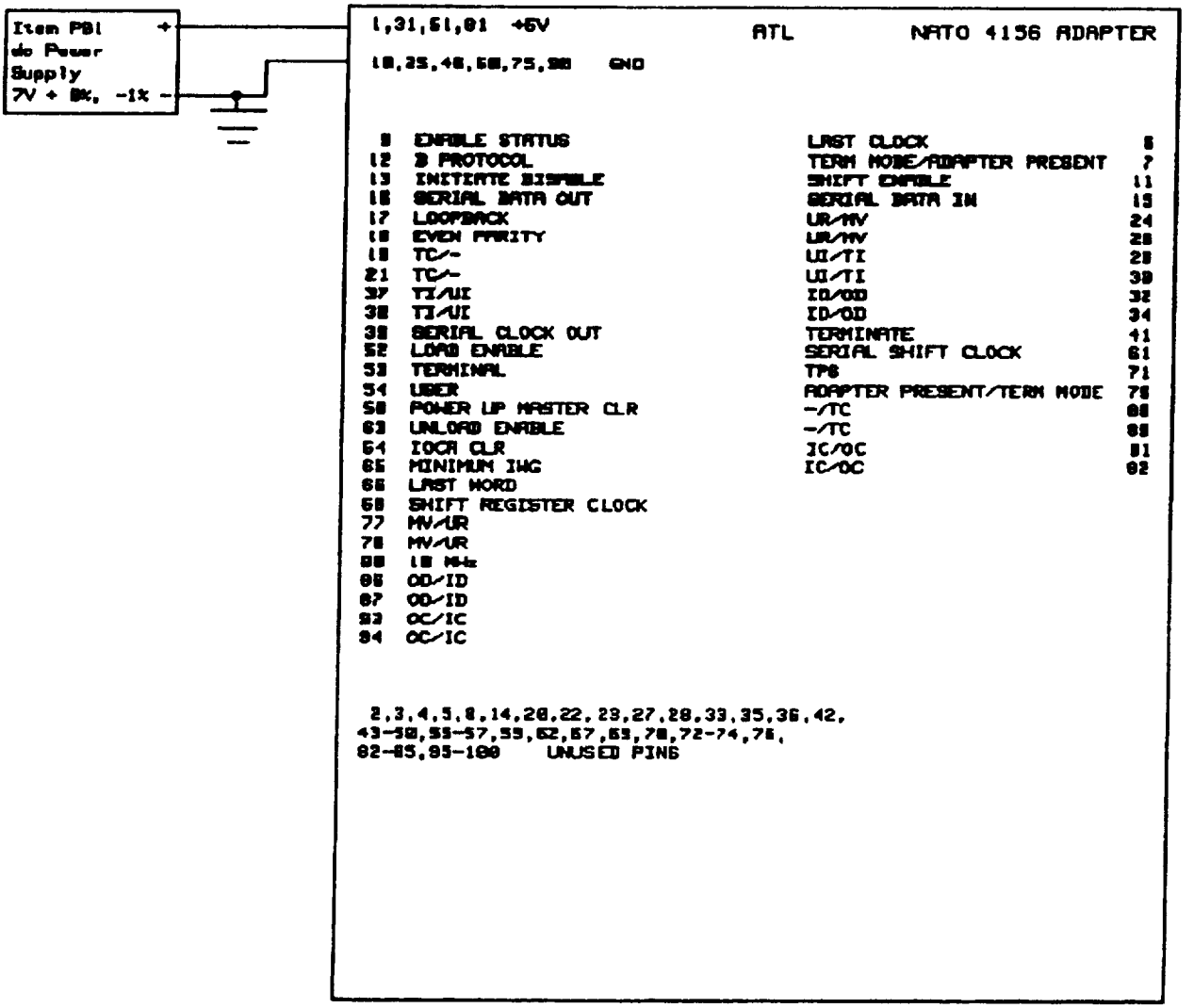


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	ENABLE S	9	5.5	-0.02	2.42	mA	1-3
D2	B PROTOCOL	12	5.5	0.0	1.1	mA	1-3
D3	INIT DISABLE	13	5.5	0.0	1.1	mA	1-3
D4	SERIAL D OUT	16	5.5	0.0	2.1	mA	1-3
D5	LOOPBACK	17	5.5	0.0	3.2	mA	1-3
D6	EVEN PARITY	18	5.5	0.0	2.0	mA	1-3
D7	SERIAL C OUT	39	5.5	0.0	3.3	mA	1-3
D8	LOAD ENABLE	52	5.5	-0.02	1.12	mA	1-3
D9	TERMINAL	53	5.5	-0.02	2.12	mA	1-3
D10	PWR UP M CLR	58	5.5	0.0	1.0	mA	1-3
D11	ULOAD ENABLE	63	5.5	-0.02	1.02	mA	1-3
D12	IOCA CLR	64	5.5	0.0	0.1	mA	1-3
D13	MINIMUM IWG	65	5.5	0.0	0.1	mA	1-3
D14	LAST WORD	66	5.5	0.0	0.2	mA	1-3
D15	SHIFT R CLK	68	5.5	0.0	1.0	mA	1-3
D16	10 MHz	80	5.5	0.0	1.0	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D16
Power supply current limit	2.61 A	NA	D1-D16
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D16
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D16 D1-D16
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 123.0 uA ± 55.0 uA ± 105.0 uA ± 160.0 uA ± 100.0 uA ± 165.0 uA ± 69.0 uA ± 108.0 uA ± 50.0 uA ± 82.5 uA ± 5.0 uA ± 10.0 uA ± 50.0 uA	D1 D2,D3 D4 D5 D6 D7 D8 D9 D10 D11 D12,D13 D14 D15,D16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E1	LAST CLOCK	6	0.0	-225.0	-50.0	mA	1-4
E2	SHIFT ENABLE	11					
E3	SERIAL D IN	15	0.0	-225.0	-50.0	mA	1-4
E4	UR/MV (L)	24	0.0	-150.0	-30.0	mA	1-4
E5	UR/MV (H)	26					
E6	UI/TI (L)	29					
E7	UI/TI (H)	30					
E8	ID/OD (L)	32					
E9	ID/OD (H)	34	0.0	-150.0	-30.0	mA	1-4
E10	TERMINATE	41	0.0	-225.0	-50.0	mA	1-4
E11	SERIAL S CLK	61	0.0	-225.0	-50.0	mA	1-4
E12	TP6	71	0.0	-150.0	-60.0	mA	1-4
E13	-/TC (L)	88	0.0	-150.0	-30.0	mA	1-4
E14	-/TC (H)	89					
E15	IC/OC (L)	91					
E16	IC/OC (H)	92	0.0	-150.0	-30.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E16
Power supply current limit	2.61 A	NA	E1-E16
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E16
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E16 E1-E16
Output current measurement	 NA NA NA NA NA	 ± 8.75 mA ± 6.00 mA ± 8.75 mA ± 4.50 mA ± 6.00 mA	 E1-E3 E4-E9 E10,E11 E12 E13-E16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	ENABLE S	9	2.4	-7.0	-0.46	mA	1-3
F2	B PROTOCOL	12	2.4	-0.85	0.05	mA	1-3
F3	INIT DISABLE	13	2.4	-0.85	0.05	mA	1-3
F4	SERIAL D OUT	16	2.4	-2.85	0.1	mA	1-3
F5	LOOPBACK	17	2.4	-4.9	0.14	mA	1-3
F6	EVEN PARITY	18	2.4	-2.25	0.08	mA	1-3
F7	TC/- (+)	19	2.4	-2.8	2.3	mA	1-3
F8	TC/- (-)	21					
F9	TI/UI (-)	37					
F10	TI/UI (+)	38	2.4	-2.8	2.3	mA	1-3
F11	SERIAL C OUT	39	2.4	-5.5	0.16	mA	1-3
F12	LOAD ENABLE	52	2.4	-2.0	-0.57	mA	1-3
F13	TERMINAL	53	2.4	-4.0	-0.52	mA	1-3
F14	USER	54	2.4	-2.04	-0.53	mA	1-3
F15	PWR UP M CLR	58	2.4	-0.25	0.03	mA	1-3
F16	ULOAD ENABLE	63	2.4	-3.15	-0.56	mA	1-3
F17	IOCA CLR	64	2.4	-0.6	0.02	mA	1-3
F18	MINIMUM IWG	65	2.4	-0.6	0.02	mA	1-3
F19	LAST WORD	66	2.4	-1.2	0.04	mA	1-3
F20	SHIFT R CLK	68	2.4	-0.25	0.03	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	MV/UR (-)	77	2.4	-2.8	2.3	mA	1-3
F22	MV/UR (+)	78	2.4	-2.8	2.3	mA	1-3
F23	10 MHz	80	2.4	-2.0	0.05	mA	1-3
F24	OD/ID (+)	86	2.4	-2.8	2.3	mA	1-3
F25	OD/ID (-)	87					
F26	OC/IC (+)	93					
F27	OC/IC (-)	94	2.4	-2.8	2.3	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F27
Power supply current limit	2.61 A	NA	F1-F27
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F27
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F27 F1-F27
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 330.0 uA ± 45.0 uA ± 147.0 uA ± 252.0 uA ± 116.0 uA ± 255.0 uA ± 283.0 uA ± 71.5 uA ± 174.0 uA ± 75.5 uA ± 14.0 uA ± 129.0 uA ± 31.0 uA ± 62.0 uA ± 14.0 uA ± 255.0 uA ± 102.0 uA ± 255.0 uA	F1 F2,F3 F4 F5 F6 F7-F10 F11 F12 F13 F14 F15 F16 F17,F18 F19 F20 F21,F22 F23 F24-F27

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	ENABLE S	9	0.4	-7.0	-0.46	mA	1-3
G2	B PROTOCOL	12	0.4	-0.85	0.05	mA	1-3
G3	INIT DISABLE	13	0.4	-0.85	0.05	mA	1-3
G4	SERIAL D OUT	16	0.4	-2.85	0.1	mA	1-3
G5	LOOPBACK	17	0.4	-4.90	0.14	mA	1-3
G6	EVEN PARITY	18	0.4	-2.25	0.08	mA	1-3
G7	TC/- (+)	19	0.4	-2.8	2.3	mA	1-3
G8	TC/- (-)	21					
G9	TI/UI (-)	37					
G10	TI/UI (+)	38	0.4	-2.8	2.3	mA	1-3
G11	SERIAL C OUT	39	0.4	-5.5	0.16	mA	1-3
G12	LOAD ENABLE	52	0.4	-2.0	-0.57	mA	1-3
G13	TERMINAL	53	0.4	-4.0	-0.52	mA	1-3
G14	USER	54	0.4	-2.04	-0.53	mA	1-3
G15	PWR UP M CLR	58	0.4	-0.25	0.03	mA	1-3
G16	ULOAD ENABLE	63	0.4	-3.15	-0.56	mA	1-3
G17	IOCA CLR	64	0.4	-0.6	0.02	mA	1-3
G18	MINIMUM IWG	65	0.4	-0.6	0.02	mA	1-3
G19	LAST WORD	66	0.4	-1.2	0.04	mA	1-3
G20	SHIFT R CLK	68	0.4	-0.25	0.03	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	MV/UR (-)	77	0.4	-2.8	2.3	mA	1-3
G22	MV/UR (+)	78	0.4	-2.8	2.3	mA	1-3
G23	10 MHz	80	0.4	-2.0	0.05	mA	1-3
G24	OD/ID (+)	86	0.4	-2.8	2.3	mA	1-3
G25	OD/ID (-)	87					
G26	OC/IC (+)	93					
G27	OC/IC (-)	94	0.4	-2.8	2.3	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G27
Power supply current limit	2.61 A	NA	G1-G27
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G27
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G27 G1-G27
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 330.0 uA ± 45.0 uA ± 147.0 uA ± 252.0 uA ± 116.0 uA ± 255.0 uA ± 283.0 uA ± 71.5 uA ± 174.0 uA ± 75.5 uA ± 14.0 uA ± 129.0 uA ± 31.0 uA ± 62.0 uA ± 14.0 uA ± 255.0 uA ± 102.0 uA ± 255.0 uA	G1 G2,G3 G4 G5 G6 G7-G10 G11 G12 G13 G14 G15 G16 G17,G18 G19 G20 G21,G22 G23 G24-G27

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	LAST CLOCK	6	-0.4	2.4	4.5	V	1-4
H2	SHIFT ENABLE	11					
H3	SERIAL D IN	15	-0.4	2.4	4.5	V	1-4
H4	UR/MV (L)	24	-0.16	2.4	4.5	V	1-4
H5	UR/MV (H)	26					
H6	UI/TI (L)	29					
H7	UI/TI (H)	30					
H8	ID/OD (L)	32					
H9	ID/OD (H)	34	-0.16	2.4	4.5	V	1-4
H10	TERMINATE	41	-0.4	2.4	4.5	V	1-4
H11	SERIAL S CLK	61					
H12	TP6	71	-0.4	2.4	4.5	V	1-4
H13	-/TC (L)	88	-0.16	2.4	4.5	V	1-4
H14	-/TC (H)	89					
H15	IC/OC (L)	91					
H16	IC/OC (H)	92	-0.16	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H16
Power supply current limit	2.61 A	NA	H1-H16
Forcing current, (Ioh)	-0.40 mA -0.16 mA -0.40 mA -0.16 mA	± 40 uA ± 16 uA ± 40 uA ± 16 uA	H1-H3 H4-H9 H10-H12 H13-H16
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H16 H1-H16
Output voltage measurement	NA	± 105 mV	H1-H16

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

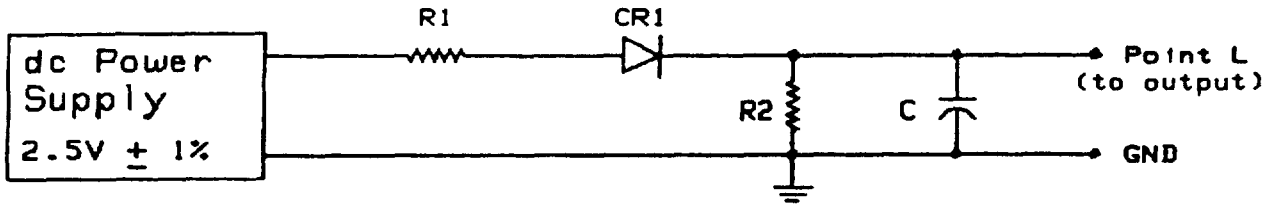
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
6,11,15,41,61,71,	6200	6.0	0.5
24,26,29,30,32,34, 88,89,91,92	16000	1.6	0.5
7,79	NA	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	LAST CLOCK	6	6.0	0.0	0.5	V	1-4
J2	TRM MD/ADP P	7	1.6	0.0	0.5	V	1-4
J3	SHIFT ENABLE	11	6.0	0.0	0.5	V	1-4
J4	SERIAL D IN	15	6.0	0.0	0.5	V	1-4
J5	UR/MV (L)	24	1.6	0.0	0.5	V	1-4
J6	UR/MV (H)	26					
J7	UI/TI (L)	29					
J8	UI/TI (H)	30					
J9	ID/OD (L)	32					
J10	ID/OD (H)	34	1.6	0.0	0.5	V	1-4
J11	TERMINATE	41	6.0	0.0	0.5	V	1-4
J12	SERIAL S CLK	61					
J13	TP6	71	6.0	0.0	0.5	V	1-4
J14	AD PRT/TRM M	79	1.6	0.0	0.5	V	1-4
J15	-/TC (L)	88					
J16	-/TC (H)	89					
J17	IC/OC (L)	91					
J18	IC/OC (H)	92	1.6	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:33Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J18
Power supply current limit	2.61 A	NA	J1-J18
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA 6.0 mA 1.6 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA ± 600 uA ± 160 uA	J1 J2 J3,J4 J5-J10 J11-J13 J14-J18
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J18 J1-J18
Output voltage measurement	NA	± 25 mV	J1-J18

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	LAST CLOCK	6	2.4	-0.05	0.05	mA	1-3
K2	TRM MD/ADP P	7	5.5	-0.65	0.30	mA	1-3
K3	SERIAL D IN	15	2.4	-0.05	0.05	mA	1-3
K4	TERMINATE	41					
K5	SERIAL S CLK	61	2.4	-0.05	0.05	mA	1-3
K6	AD PRT/TRM M	79	5.5	-0.85	0.30	mA	1-3
Low level							
K7	LAST CLOCK	6	0.4	-0.05	0.05	mA	1-3
K8	SERIAL D IN	15					
K9	TERMINATE	41					
K10	SERIAL S CLK	61	0.4	-0.05	0.05	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K10
Power supply current limit	2.61 A	NA	K1-K10
Forcing voltage,			
(Voh)	2.4 V	± 50 mV	K1
(Voh)	5.5 V	± 50 mV	K2
(Voh)	2.4 V	± 50 mV	K3-K5
(Voh)	5.5 V	± 50 mV	K6
(Vol)	0.4 V	± 50 mV	K7-K10
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	K1-K10
low, (VL)	0.0 V	± 100 mV	K1-K10
Output current measurement	NA	± 5 uA	K1
	NA	± 5 uA	K2
	NA	± 5 uA	K3-K5
	NA	± 6 uA	K6
	NA	± 5 uA	K7-K10

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	B PROTOCOL	12	-18.0	-1.5	0.0	V	1-3
L2	INIT DISABLE	13	-18.0	-1.5	0.0	V	1-3
L3	SERIAL D OUT	16	-18.0	-1.2	0.0	V	1-3
L4	LOOPBACK	17	"	"	"	"	"
L5	EVEN PARITY	18	"	"	"	"	"
L6	SERIAL C OUT	39	-18.0	-1.2	0.0	V	1-3
L7	PWR UP M CLR	58	-18.0	-1.5	0.0	V	1-3
L8	IOCA CLR	64	-18.0	-1.2	0.0	V	1-3
L9	MINIMUM IWG	65	"	"	"	"	"
L10	LAST WORD	66	-18.0	-1.2	0.0	V	1-3
L11	SHIFT R CLK	68	-18.0	-1.5	0.0	V	1-3
L12	10 MHz	80	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L12
Power supply current limit	NA	NA	L1-L12
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L12
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L12 L1-L12
Input voltage measurement	NA NA NA NA NA NA	± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV	L1,L2 L3-L6 L7 L8-L10 L11 L12

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	8.4	15.6	uF	1-4
M2	PSFC	51	5.6	10.4	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 360 nF ± 240 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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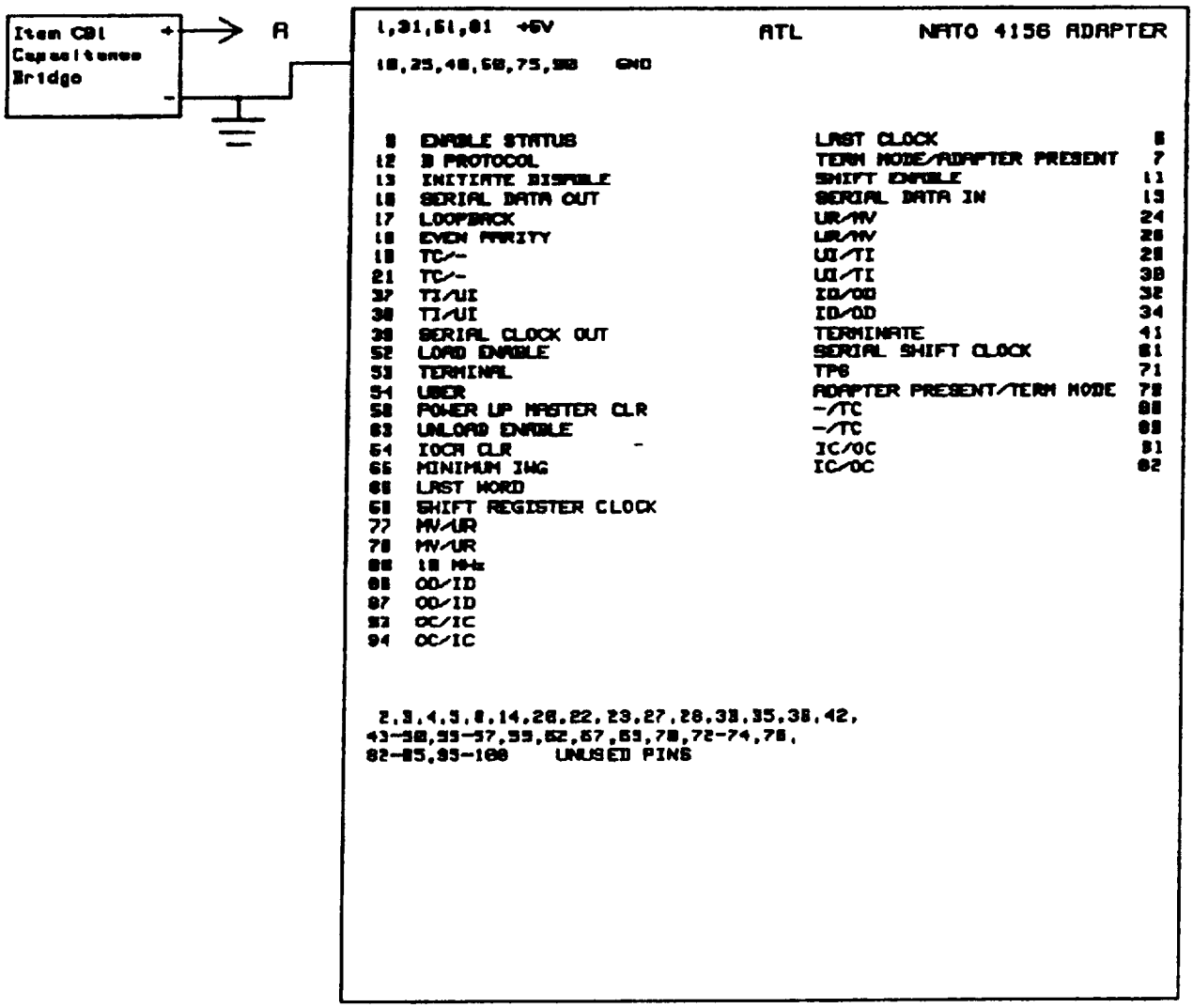


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	725	1740	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.61 A	NA	N
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	N N
Power supply current measurement	NA	± 51 mA	N

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2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (425 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (425 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.61 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (425 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (425 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (425 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (425 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.61 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

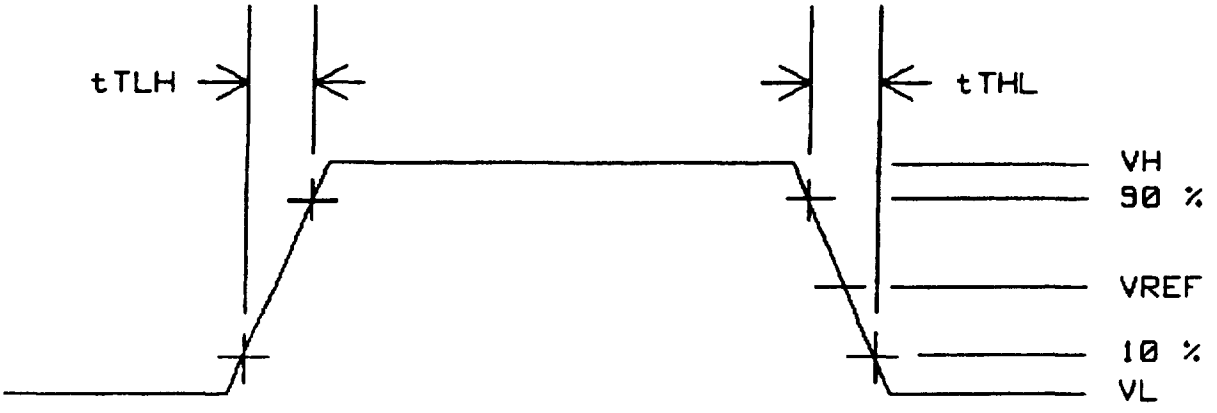
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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Table XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	OC/IC (+) to SHIFT ENABLE (low level)	-	-	-	212	212	212	nS	11
1	tpd	OC/IC (-) to SHIFT ENABLE (low level)	-	-	-	212	212	212	nS	11
2	tpd	TI/UI (-) to SHIFT ENABLE (high level)	-	-	-	327	327	327	nS	11
2	tpd	TI/UI (+) to SHIFT ENABLE (high level)	-	-	-	327	327	327	nS	11
3	tpd	MV/UR (-) to TERMINATE	-	-	-	256	256	256	nS	11
3	tpd	MV/UR (+) to TERMINATE	-	-	-	256	256	256	nS	11
4	tpd	OC/IC (+) to SERIAL S CLK	-	-	-	109	109	109	nS	11
4	tpd	OC/IC (-)to SERIAL S CLK	-	-	-	109	109	109	nS	11
5	tpd	OD/ID (+) to SERIAL D IN	-	-	-	127	127	127	nS	11
5	tpd	OD/ID (-) to SERIAL D IN	-	-	-	127	127	127	nS	11
6	tpd	TC/- (+) to LAST CLOCK	-	-	-	127	127	127	nS	11
6	tpd	TC/- (-) to LAST CLOCK	-	-	-	127	127	127	nS	11
7	tpd	MV/UR (-) to UI/TI (L)	-	-	-	211	211	211	nS	11
7	tpd	MV/UR (-) to UI/TI (H)	-	-	-	211	211	211	nS	11
7	tpd	MV/UR (+) to UI/TI (L)	-	-	-	211	211	211	nS	11
7	tpd	MV/UR (+) to UI/TI (H)	-	-	-	211	211	211	nS	11

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Table XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	ULOAD ENABLE to UR/MV (L)	-	-	-	131	131	131	nS	11
8	tpd	ULOAD ENABLE to UR/MV (H)	-	-	-	131	131	131	nS	11
9	tpd	SHIFT R CLK to IC/OC (L)	-	-	-	94	94	94	nS	11
9	tpd	SHIFT R CLK to IC/OC (H)	-	-	-	94	94	94	nS	11
10	tpd	SERIAL C OUT to -/TC (L)	-	-	-	82	82	82	nS	11
10	tpd	SERIAL C OUT to -/TC (H)	-	-	-	82	82	82	nS	11

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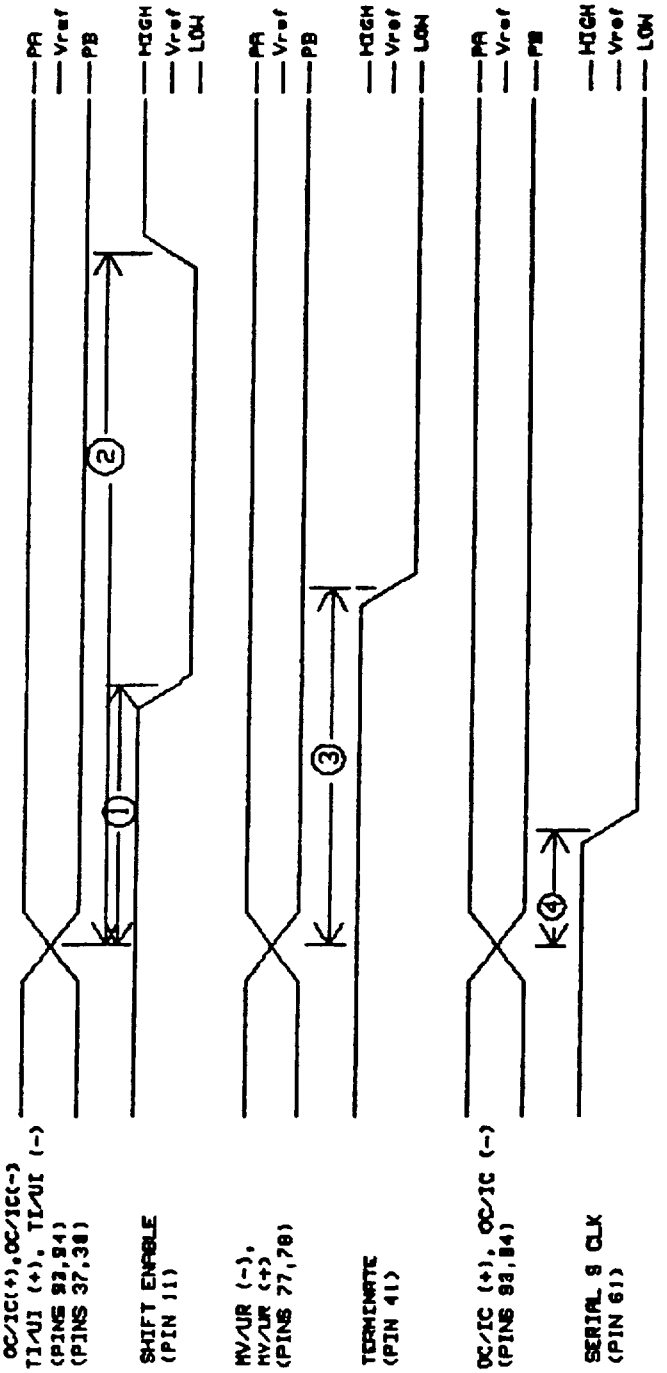


FIGURE 11a. Dynamic waveforms for table XVII.

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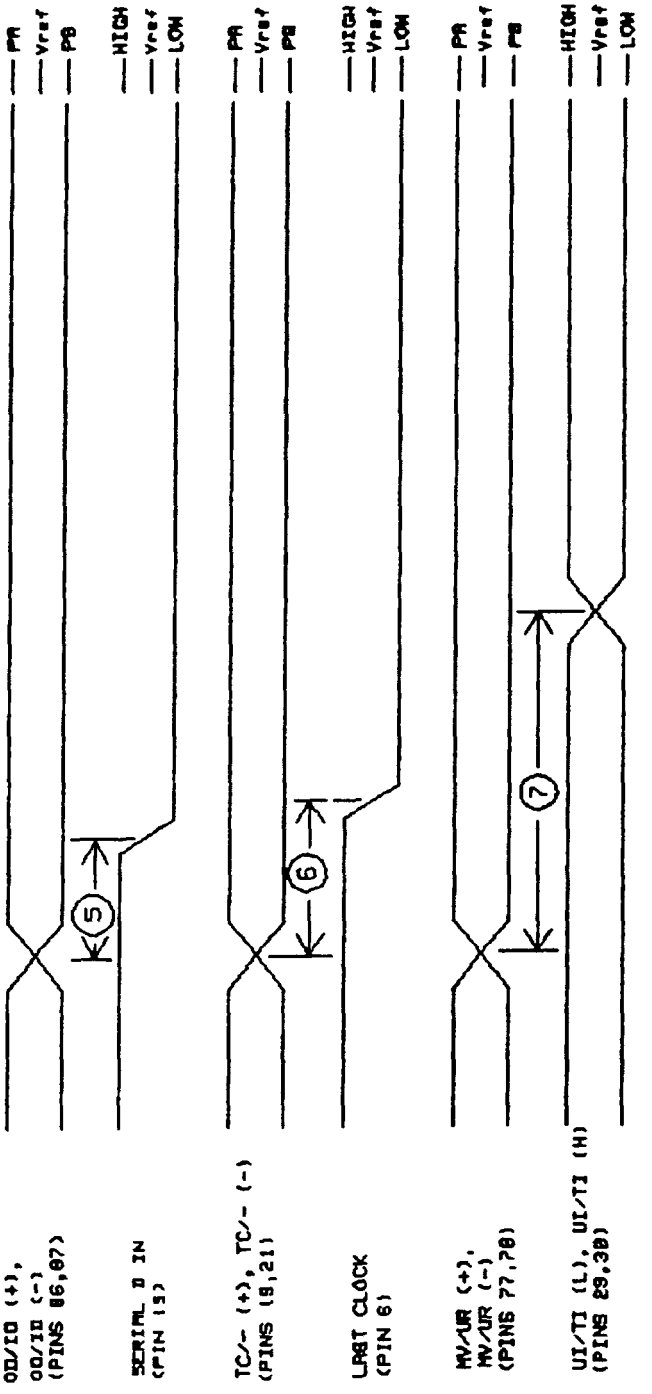


FIGURE 11b. Dynamic waveforms for table XVII.

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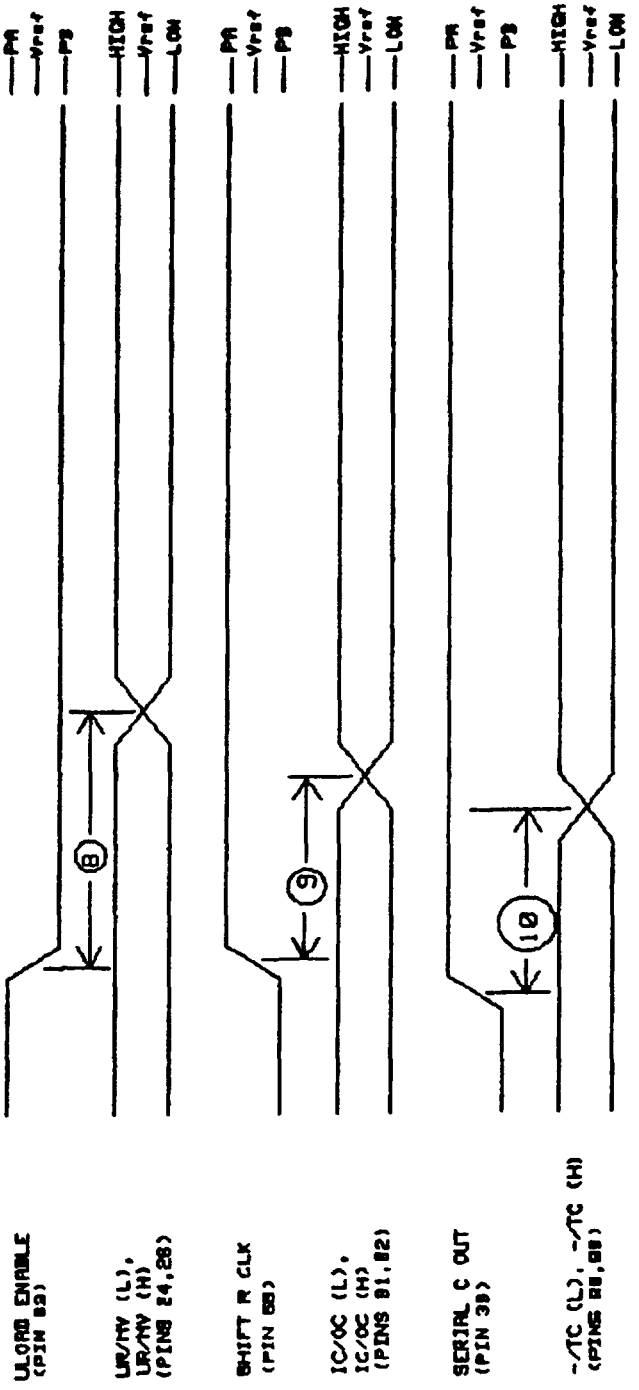


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R1	ENABLE S	9	14	200	pF	1-4
R2	B PROTOCOL	12	4	175	pF	1-4
R3	INIT DISABLE	13	4	150	pF	1-4
R4	SERIAL D OUT	16	6	125	pF	1-4
R5	LOOPBACK	17	10	200	pF	1-4
R6	EVEN PARITY	18	4	100	pF	1-4
R7	TC/- (+)	19	2	100	pF	1-4
R8	TC/- (-)	21				
R9	TI/UI (-)	37	2	100	pF	1-4
R10	TI/UI (+)	38	2	75	pF	1-4
R11	SERIAL C OUT	39	12	175	pF	1-4
R12	LOAD ENABLE	52	2	125	pF	1-4
R13	TERMINAL	53	6	125	pF	1-4
R14	USER	54	9	200	pF	1-4
R15	PWR UP M CLR	58	2	125	pF	1-4
R16	ULOAD ENABLE	63	6	100	pF	1-4
R17	IOCA CLR	64	2	75	pF	1-4
R18	MINIMUM IWG	65	2	75	pF	1-4
R19	LAST WORD	66	2	100	pF	1-4
R20	SHIFT R CLK	68	2	100	pF	1-4

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R21	MV/UR (-)	77	2	75	pF	1-4
R22	MV/UR (+)	78				
R23	10 MHz	80				
R24	OD/ID (+)	86				
R25	OD/ID (-)	87				
R26	OC/IC (+)	93				
R27	OC/IC (-)	94	2	75	pF	1-4

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NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R27
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R27
Capacitance measurement	NA	± 9.3 pF	R1
	NA	± 8.55 pF	R2
	NA	± 7.3 pF	R3
	NA	± 5.95 pF	R4
	NA	± 9.5 pF	R5
	NA	± 4.8 pF	R6
	NA	± 4.9 pF	R7-R9
	NA	± 3.65 pF	R10
	NA	± 8.15 pF	R11
	NA	± 6.15 pF	R12
	NA	± 5.95 pF	R13
	NA	± 9.55 pF	R14
	NA	± 6.15 pF	R15
	NA	± 4.7 pF	R16
	NA	± 3.65 pF	R17,R18
	NA	± 4.9 pF	R19,R20
	NA	± 3.65 pF	R21-R27

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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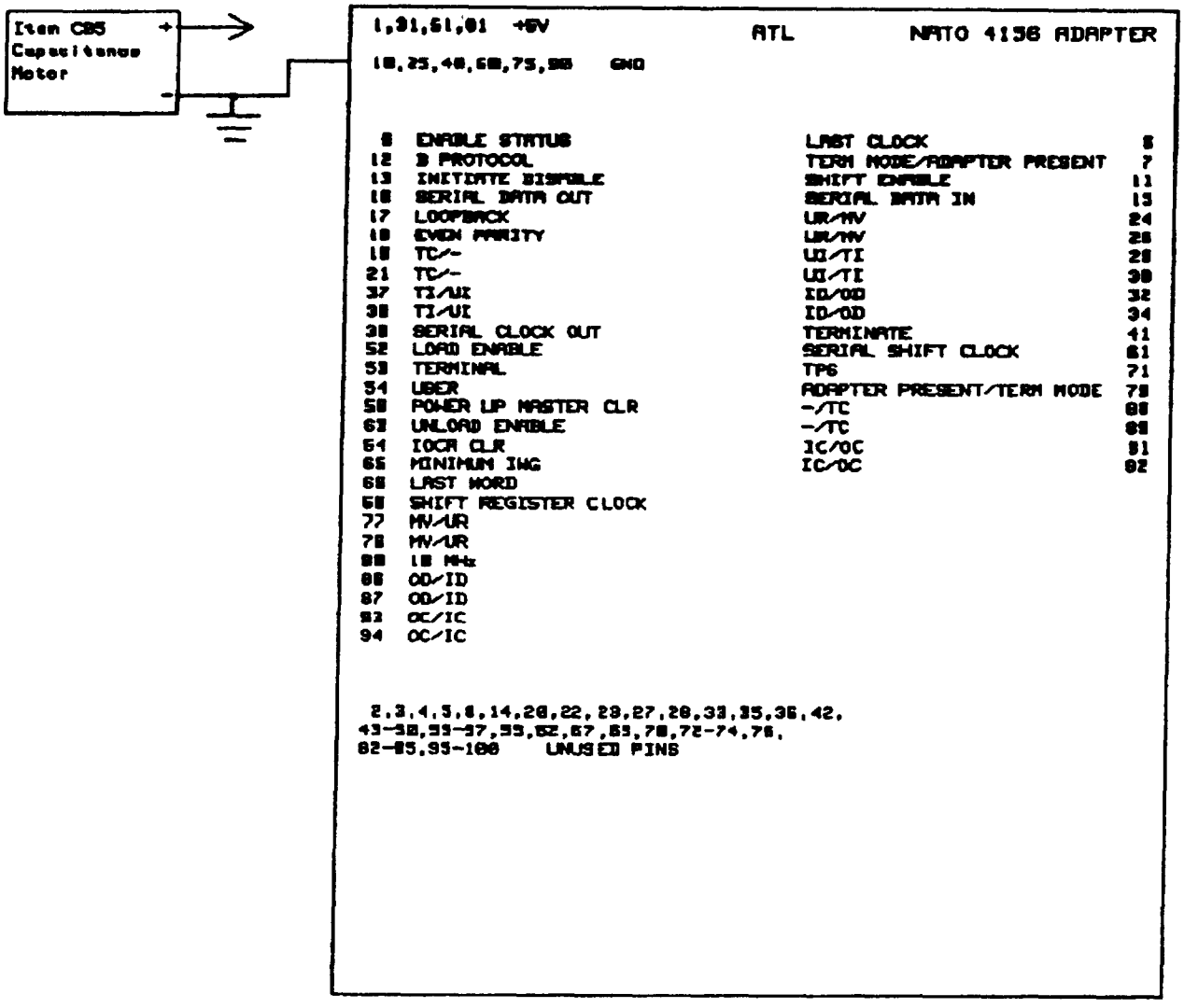


FIGURE 12. Input capacitance test circuit for table XVIII.

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TABLE XIX. Input clamp diode latchup test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
S	SERIAL C OUT	39	-18.0	-1.2	2.4	V	1-3

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NOTES:

1. Input clamp diode latchup test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5	± 45 mV	S
Power supply current limit	2.61 A	NA	S
Forcing current, (Iin)	-18 mA	± 1.8 mA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Input voltage measurement	NA	± 180 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIX and the voltage then measured.

(d) Apply pulse to the module for a duration of 500 ± 20 nS.

3. Abbreviations used are defined in 6.1.

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TABLE XX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	2.61 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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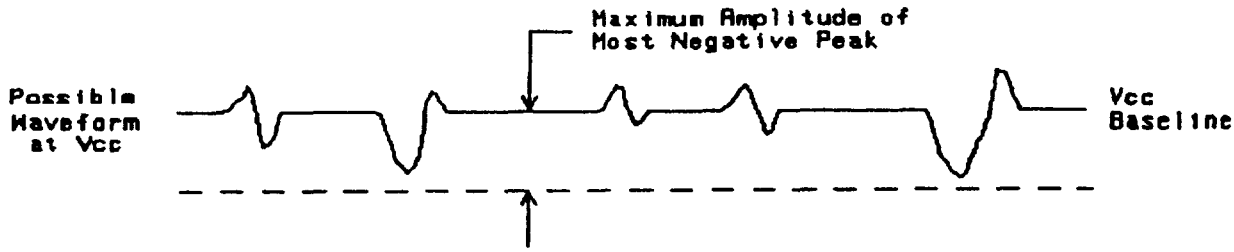
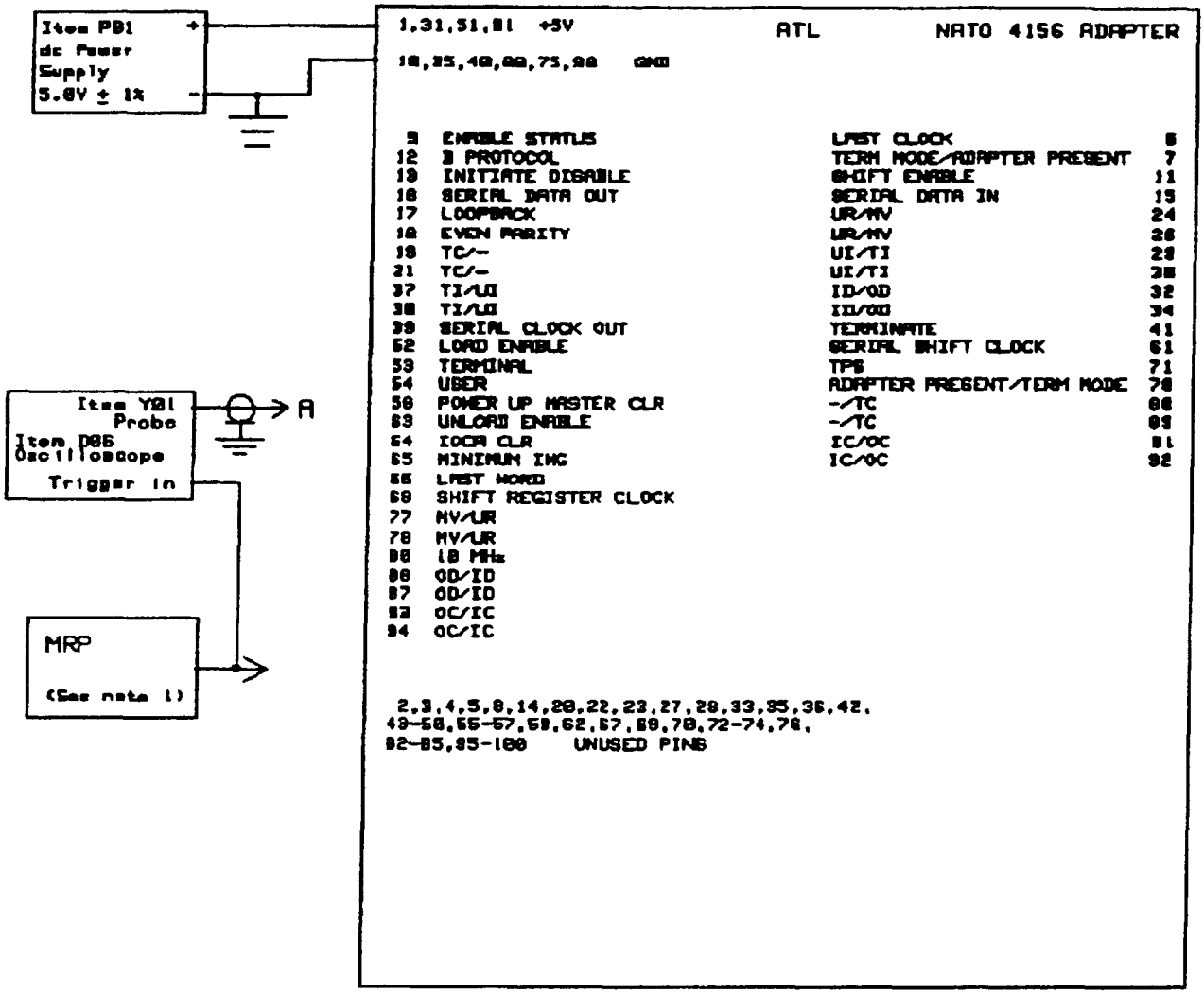


FIGURE 13. Power supply transient amplitude waveform for table XX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power suppply transient amplitude test circuit for table XX.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-28787/524	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE Modules, Standard Electronic, NATO-STD-4156 (Modified) Channel Adapter Module Key Code ATL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)

7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: TPOC: 812-854-1528 (2) AUTOVON: 88-482-1528
c. ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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|INCH-POUND|

MIL-M-28787/524
AMENDMENT 1
29 September 1993

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

NATO-STD-4156 (MODIFIED) CHANNEL ADAPTER MODULE, KEY CODE ATL

This amendment forms a part of Military Specification MIL-M-28787/524, dated 26 November 1991, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I, Test conditions A1 and A2-A46: Add End-of-life limits below each line with a minimum limit of 1 MOhm (all temperatures) and a maximum limit of INF MOhm (all temperatures).

PAGE 7

TABLE I, Test condition M1, Min column (all temperatures): Delete "8.4" (three places) and substitute "4.2" (three places).

TABLE I, Test condition M1, Max column (all temperatures): Delete "15.6" (three places) and substitute "7.8" (three places).

TABLE I, Test condition M2, Min column (all temperatures): Delete "5.6" (three places) and substitute "4.2" (three places).

TABLE I, Test condition M2, Max column (all temperatures): Delete "10.4" (three places) and substitute "7.8" (three places).

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